Prop

Let X be any space and Y be Compact Hausdorff space.

$f: X \to Y$ is Continuous map if and only if the graph $G_f = \{(x, f(x)) \mid x \in X\}$ is closed.

$\xrightarrow{T_2}$ $\xleftarrow{\text{compact}}$

proof. Let $f: X \to Y$ be a Continuous map.

Let $(x,y) \in X \times Y \setminus G_f$ be given. Then, $y \neq f(x)$, so there exist disjoint open sets $\mathcal{U}, V$

$$\mathcal{U} \cap V = \phi, \quad y \in \mathcal{U}, \quad f(x) \in V$$

Since $f(x) \in V \overset{\text{def}}{\iff} x \in f^{-1}[V]$ and f is Continuous, $f^{-1}[V]$ is an open set in X.

Now, $(x,y) \in f^{-1}[V] \times \mathcal{U}$

And, $(f^{-1}[V] \times \mathcal{U}) \cap G_f = \phi$, because if $(x, f(x)) \in f^{-1}[V] \times \mathcal{U}$, then

$f(x) \in V$ and $f(x) \in \mathcal{U}$, it is contradiction to disjointness.

Therefore, $(x,y) \in f^{-1}[V] \times \mathcal{U} \subseteq X \times Y \setminus G_f$, So proved. $\rfloor \Rightarrow$

Conversely, let G_f is closed in $X \times Y$, Under the Product Topology.

Let $C \subseteq Y$ be a Closed set in Y. Then, $G_f \cap (X \times C)$ is closed. Moreover,

$$P_X[G_f \cap (X \times C)] = \{x \in X \mid f(x) \in C\} = f^{-1}[C]$$

is closed, because $P_X : X \times Y \to X : (x,y) \mapsto x$ is closed map if Y is Compact. $\rfloor$ ☒ $\Leftarrow$

Note: for $g = f \times i : X \times Y \to Y \times Y : (x,y) \mapsto (f(x), y)$,

$$g^{-1}[\Delta(Y)] = \{(x,y) \in X \times Y \mid (f(x), y) \in \Delta(Y)\} = G_f$$

「Prop.

Compact Hausdorff space is Normal space.